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# Exploring the complex associations between community public spaces and healthy aging: an explainable analysis using catboost and SHAP

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## Abstract

**Background** As global aging accelerates, community public spaces (CPS) are increasingly recognized as vital for promoting healthy aging. However, existing research often employs linear analytical methods or focuses on single health dimensions, overlooking how CPS features are non-linearly and interactively related to multiple aspects of elderly well-being.

**Methods** This study analyzed data from 2,508 older adults in Shanghai to examine the associations between 13 CPS environmental features and physical, mental, and social health. An explainable machine learning approach, incorporating CatBoost and SHAP, was used to identify key correlates and patterns of interaction among features.

**Results** Destination accessibility, public transport, residential density, and walking infrastructure were positively associated with all health dimensions, while traffic complexity and crime rate showed consistent negative associations. SHAP interaction analysis revealed that health benefits often emerge from synergistic combinations of features, rather than from the effects of individual features in isolation.

**Conclusions** This study provides new insights into the multidimensional and non-linear relationships between CPS features and elderly health. The findings offer empirical evidence to guide community planning and intervention strategies that foster health-supportive environments for aging populations.

**Keywords** Community public space, Healthy aging, CatBoost, SHAP analysis, Elderly health, Explainable machine learning

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## Background

Aging is often accompanied by increased susceptibility to chronic diseases, cognitive decline, and mental health challenges, such as depression [1]. These issues not only diminish the quality of life for older adults but also impose substantial burdens on healthcare systems worldwide. Consequently, promoting healthy aging has become a critical societal and scientific priority. Community public spaces are increasingly recognized as essential resources for supporting the health and well-being of older adults. These environments provide opportunities for physical activity, social interaction, and emotional support, all of which are vital for maintaining functional capacity and fostering independence [2, 3]. Empirical evidence has demonstrated that well-designed public spaces can alleviate loneliness, improve mental health, and enhance overall quality of life in aging populations [4, 5]. As the demand for age-friendly communities grows, understanding the specific pathways through which community environments are associated with multidimensional health is critical for guiding evidence-based urban planning.

Physical activity, a cornerstone of healthy aging, is profoundly shaped by community environments. Walkable neighborhoods, characterized by features such as road connectivity, diverse land use, and proximity to amenities, have been shown to significantly enhance physical activity levels, thereby reducing the risk of hypertension, diabetes, and cardiovascular diseases [6, 7]. Green spaces, in particular, provide restorative environments for low-intensity exercise, especially for individuals with limited mobility, and have been associated with improved physical functioning and reduced risks of chronic conditions [5, 8]. Furthermore, the quality of built environments—encompassing pedestrian infrastructure, safety measures, and aesthetic attributes—has been identified as a critical determinant of physical activity and mobility in older adults, influencing outcomes such as obesity prevention and cardiac health maintenance [9, 10]. These findings underscore the integral role of community environments in sustaining physical health and preventing functional decline among aging populations.

Beyond physical health, the design and quality of community spaces are closely associated with mental well-being. Accessible, aesthetically pleasing public spaces are associated with reduced symptoms of depression and anxiety, as they provide opportunities for relaxation, social engagement, and stress reduction [11]. Recreational facilities that promote informal gatherings further enhance emotional resilience and foster a sense of belonging [12, 13]. Moreover, walkable and safe environments mitigate loneliness, a prevalent risk factor for mental health disorders in older adults, by facilitating social connections and encouraging active participation

in community life [14, 15]. These findings suggest that the strategic design of public spaces can serve as an important intervention for improving mental health outcomes in aging populations.

Social health, encompassing social participation and connectedness, is another critical dimension associated with community public spaces. Proximity to accessible public spaces, such as parks and community centers, has been consistently linked to greater levels of social interaction and participation among older adults [16, 17]. Inclusive and barrier-free designs are particularly important for fostering social integration, as they address mobility challenges and reduce perceived isolation [4, 18]. Furthermore, intergenerational spaces, such as community gardens and shared recreational areas, facilitate meaningful social interactions and contribute to the development of cohesive, age-friendly communities [19]. These findings highlight the importance of designing environments that promote both individual well-being and broader social inclusion, particularly for aging populations vulnerable to social isolation.

While substantial research has explored the association of community public spaces on physical, mental, and social health, much of the existing literature tends to treat these dimensions separately. Environmental factors are often examined in isolation, with different predictors prioritized for each health outcome. Moreover, few studies have examined how these environmental variables interact with one another, or whether their associations vary across different health domains. Most importantly, the vast majority of existing work relies on traditional linear models, which are limited in capturing the complex, nonlinear, and interaction-based mechanisms that characterize the relationship between community environments and elderly health. These limitations underscore the need for a more integrated, interpretable, and multidimensional analytical approach.

In response to these analytical challenges, recent research has increasingly adopted Explainable Artificial Intelligence (XAI) methods. XAI techniques offer a promising way to improve the transparency and interpretability of complex machine learning models, enabling researchers to derive more meaningful insights for empirical understanding and practical decision-making [20]. Among them, SHAP (SHapley Additive exPlanations) has gained particular traction due to its solid theoretical foundation in cooperative game theory and its ability to provide additive importance values for individual predictions [21]. SHAP not only identifies the most relevant features but also clarifies their contribution patterns and interaction dynamics, making it especially suitable for studies aiming to unpack multidimensional health outcomes.

Building on this progress, the present study proposes an integrated analytical framework combining CatBoost and SHAP. First, it simultaneously models three key health dimensions—physical, mental, and social—within a unified machine learning system, enabling a holistic assessment. Second, by leveraging CatBoost’s capacity to model nonlinear patterns and SHAP’s interpretability, the framework captures complex relationships that conventional methods would overlook. In particular, we adopt both global analysis, which identifies the overall importance and direction of each feature across the full sample, and local analysis, which explores pairwise interactions to uncover nuanced, context-dependent mechanisms. This dual approach enables a more comprehensive understanding of the patterns linking environmental features with elderly health at both population and individual levels. Third, this framework yields interpretable

insights that can directly inform age-friendly urban design and targeted public health interventions. Together, these innovations enhance both methodological rigor and practical relevance in the study of aging and the built environment.

Specifically, this study seeks to address three key questions: [1] Which community public space features are most critical for influencing older adults’ health across physical, mental, and social dimensions [2]? Do these environmental features exert distinct associations across these health domains [3]? How do these features interact, and what are the implications for promoting healthy aging? By answering these questions, the study aims to uncover the mechanisms through which community environments shape multidimensional health outcomes and provide evidence-based recommendations for designing age-friendly public spaces that enhance older adults’ well-being.

**Table 1** Descriptive statistics of demographic and geographic characteristics of participants

| Variable                 | Category                        | Count | Percentage (%) |
|--------------------------|---------------------------------|-------|----------------|
| gender                   | Male                            | 1263  | 50.36          |
|                          | Female                          | 1245  | 49.64          |
| age                      | 60–69                           | 544   | 21.69          |
|                          | 70–79                           | 725   | 28.91          |
|                          | 80–89                           | 694   | 27.67          |
|                          | 90+                             | 545   | 21.73          |
|                          |                                 |       |                |
| marriage                 | Single                          | 197   | 7.85           |
|                          | Married                         | 1249  | 49.8           |
|                          | Divorced                        | 290   | 11.56          |
|                          | Widowed                         | 772   | 30.79          |
| education                | Primary or below                | 1789  | 71.33          |
|                          | Middle School                   | 415   | 16.55          |
|                          | High School/Technical           | 224   | 8.93           |
|                          | Associate Degree                | 52    | 2.07           |
|                          | Bachelor’s or above             | 28    | 1.12           |
| income                   | ≤ 3000 RMB                      | 1976  | 78.79          |
|                          | 3001–5000 RMB                   | 284   | 11.32          |
|                          | 5001–8000 RMB                   | 87    | 3.47           |
|                          | ≥ 8001 RMB                      | 83    | 3.31           |
|                          | Prefer not to disclose          | 78    | 3.11           |
| disease                  | No chronic disease              | 971   | 38.72          |
|                          | One chronic disease             | 677   | 26.99          |
|                          | Two or more chronic diseases    | 860   | 34.29          |
| District<br>(Geographic) | Hongkou (Inner Ring)            | 385   | 15.4           |
|                          | Putuo (Inner–Middle Ring)       | 362   | 14.4           |
|                          | Yangpu (Inner–Middle Ring)      | 371   | 14.8           |
|                          | Pudong (Mixed Inner–Outer Ring) | 428   | 17.1           |
|                          | Jiading (Outer Ring)            | 330   | 13.2           |
|                          | Songjiang (Outer Ring)          | 320   | 12.8           |
|                          | Qingpu (Outer Ring)             | 312   | 12.4           |

Methods

Data and sample

Data for this study were collected between May and August 2024 through structured questionnaires administered in collaboration with district-level government offices and local resident committees. A total of 2,508 permanent residents aged 60 years and above participated in the survey. To ensure broad representativeness across Shanghai’s diverse urban landscape, participants were recruited from seven administrative districts: Hongkou, Putuo, Yangpu, Jiading, Qingpu, Pudong, and Songjiang. These districts span the city’s inner, middle, and outer ring areas, capturing a wide spectrum of community types from dense urban centers to suburban and peri-urban settings.

Participants were selected through a stratified community-based sampling approach. With the assistance of neighborhood committees, eligible older adults were invited to participate based on age and residence criteria. To ensure data quality given the length and complexity of the questionnaire, respondents were screened for cognitive ability and provided with guided support where necessary. The structured recruitment process helped ensure both the validity of responses and the demographic diversity of the sample.

The questionnaire included validated items covering built environment features, health status, and sociodemographic characteristics. Ethical approval was obtained from the institutional Ethics Committee, and all participants provided informed consent after being briefed on the study’s objectives, procedures, and confidentiality safeguards. Table 1 summarizes the demographic characteristics and geographic distribution of the sample. To enhance spatial contextualization, each administrative district was additionally categorized by its predominant

position within Shanghai's three-ring urban structure (inner, middle, and outer rings), a system widely used in local urban planning.

### Measures

All independent, dependent, and control variables were collected concurrently through the same questionnaire between May and August 2024. To operationalize the 13 community public space (CPS) variables, we employed validated instruments and multiple-item Likert-type scales. Appendix A provides detailed information on the questionnaire items and their corresponding references for each variable. The independent variables, representing community public space factors, were derived from the literature (Table 2) and assessed using validated scales.

X1–X8 were adapted from the Chinese version of the Neighborhood Environment Walkability Scale—Abbreviated (NEWS-A), a validated instrument widely used in aging and mobility research [22], with response options ranging from 1 (strongly disagree/short time) to 4 or 5 (strongly agree/long time), depending on the item format.

X9–X11 were based on the perceived neighborhood environment scale [23], including perceptions of neighborhood infrastructure availability and accessibility, with three to four items each. Response formats also followed a 4-point Likert scale.

X12 was measured using five items covering abundance, visibility, access, usage, and quality of green space, adapted from the measures of perceived green space exposure [11]. Each item used a 5-point agreement scale to reflect the subjective exposure and engagement with greenery in daily life.

X13 was derived from the built environment quality scale [3], comprised four items measuring perceived safety, cleanliness, air pollution, and noise, using a 5-point agreement scale to capture environmental satisfaction and discomfort.

All independent variables were measured using established instruments listed in Appendix A. Most indicators were calculated as mean scores of their respective items, while a few followed specific scoring rules or included reverse-coded items, as detailed in the original references. Full scoring procedures are available in the cited source materials.

Dependent variables—physical health (PH), mental health (MH), and social health (SH)—were evaluated using the SF-36 Health Survey, a well-established tool for assessing health across multiple dimensions [40]. Following the official scoring guidelines, we derived the three composite health outcomes based on specific subscales: PH was computed using the domains of Physical Functioning, Role-Physical, and Bodily Pain; MH was constructed from Mental Health, Role-Emotional, and Vitality; and SH was represented by the Social

**Table 2** Environmental features of community public spaces affecting older adults' health

| No  | Variable                  | Interpretation                                                                                                                                                                                     | References   |
|-----|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| X1  | Residential density       | The overall concentration of housing types within a neighborhood, from detached homes to multi-story buildings.                                                                                    | [24–26]      |
| X2  | Land-use diversity        | The variety and accessibility of different land uses, showing the diversity and availability of local amenities.                                                                                   | [18, 27, 28] |
| X3  | Destination accessibility | The ease of access to amenities within walking distance, indicating neighborhood walkability.                                                                                                      |              |
| X4  | Street connectivity       | The frequency of intersections and availability of alternative routes, allowing for ease of movement and connectivity.                                                                             | [29, 30]     |
| X5  | Walking infrastructure    | The presence of sidewalks, street lighting, pedestrian and cyclist visibility, and crosswalks and pedestrian signals, supporting a walkable environment.                                           | [31–33]      |
| X6  | Aesthetics                | The presence of street trees, visually engaging elements, attractive natural features, and appealing architecture, contributing to neighborhood visual quality and attractiveness.                 | [34, 35]     |
| X7  | Traffic complexity        | Traffic conditions—including volume, speed, and driver behavior—are closely related to pedestrian safety and comfort.                                                                              | [7, 10, 36]  |
| X8  | Crime rate                | Residents' perceptions of neighborhood crime rates and their associations with perceived walking safety during the day and night.                                                                  | [31, 37, 38] |
| X9  | Public transport          | The availability and accessibility of bus and subway stations within a neighborhood, indicating ease of access to public transportation options.                                                   | [6, 8]       |
| X10 | Outdoor public spaces     | The availability and accessibility of parks, entertainment squares, and long corridors within a neighborhood, providing open areas for recreation, social interaction, and physical activity.      | [13, 19, 35] |
| X11 | Barrier-free design       | The presence of barrier-free features at residential entrances, in public spaces, and community facilities, ensuring accessibility for individuals with mobility needs.                            | [5]          |
| X12 | Green space               | The abundance, visibility, accessibility, usage, and quality of natural vegetation areas within a neighborhood.                                                                                    | [11, 19, 32] |
| X13 | Built environment quality | The overall safety, cleanliness of public spaces, air quality, and noise levels within a neighborhood, reflecting environmental quality and its potential relationship with residents' well-being. | [1, 3, 39]   |

Functioning dimension. The General Health domain was excluded from aggregation to avoid conceptual overlap, and the Health Transition item was also omitted from scoring, as it serves only as a retrospective self-assessment. Each dimension was calculated following standard SF-36 procedures.

Control variables included age, gender, education level, marital status, chronic disease status, and income, as these factors have been shown to be associated with health outcomes in older adults [2, 33, 41].

### Analysis

CatBoost, a gradient-boosting algorithm, was employed to model the relationships between community and demographic features and health outcomes. Compared to traditional gradient boosting frameworks, CatBoost is optimized for handling categorical variables, reduces overfitting through ordered boosting, and shows strong performance on small-to-medium-sized datasets without extensive preprocessing [42]. The model's objective function is defined as:

$$\mathcal{L} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (1)$$

where  $l(y_i, \hat{y}_i)$  represents the loss function quantifying prediction error, and  $\Omega(f_k)$  is the regularization term controlling model complexity.

To interpret CatBoost predictions, SHAP (Shapley Additive Explanations) values were calculated. SHAP assigns an importance score to each feature by quantifying its marginal contribution to model predictions, based on Shapley values from cooperative game theory [43]:

$$\varphi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)] \quad (2)$$

where  $S$  is a subset of features excluding  $i$ , and  $v(S)$  represents the model output for subset  $S$ .

SHAP values provided insights into the relative importance of community features, their interactions, and their contributions on each health dimension. By analyzing these values, this study identified key predictors and potential synergies among features, enhancing the transparency and interpretability of model predictions.

### Model comparison and selection

To ensure the reliability and accuracy of the prediction models, we conducted a comparative evaluation of five machine learning algorithms—Linear Regression, Random Forest, LightGBM, XGBoost, and CatBoost—across three health outcomes (physical, mental, and

**Table 3** Model comparison

| Target | Model            | R <sup>2</sup> | MSE   | MAE   |
|--------|------------------|----------------|-------|-------|
| PH     | LinearRegression | 0.727          | 0.063 | 0.164 |
|        | RandomForest     | 0.746          | 0.058 | 0.142 |
|        | LightGBM         | 0.745          | 0.059 | 0.143 |
|        | XGBoost          | 0.737          | 0.060 | 0.145 |
|        | CatBoost         | 0.755          | 0.056 | 0.143 |
| MH     | LinearRegression | 0.685          | 0.021 | 0.095 |
|        | RandomForest     | 0.712          | 0.019 | 0.082 |
|        | LightGBM         | 0.724          | 0.018 | 0.076 |
|        | XGBoost          | 0.731          | 0.018 | 0.077 |
|        | CatBoost         | 0.731          | 0.018 | 0.076 |
| SH     | LinearRegression | 0.594          | 0.011 | 0.080 |
|        | RandomForest     | 0.604          | 0.010 | 0.079 |
|        | LightGBM         | 0.592          | 0.011 | 0.081 |
|        | XGBoost          | 0.602          | 0.011 | 0.080 |
|        | CatBoost         | 0.606          | 0.010 | 0.079 |

**Table 4** Sensitivity analysis

| Target | Random Seed | R <sup>2</sup> | MSE   |
|--------|-------------|----------------|-------|
| PH     | 42          | 0.755          | 0.056 |
| MH     | 42          | 0.731          | 0.018 |
| SH     | 42          | 0.606          | 0.010 |
| PH     | 77          | 0.635          | 0.083 |
| MH     | 77          | 0.649          | 0.024 |
| SH     | 77          | 0.574          | 0.012 |
| PH     | 2024        | 0.675          | 0.078 |
| MH     | 2024        | 0.690          | 0.023 |
| SH     | 2024        | 0.555          | 0.012 |

social health). Tenfold cross-validation was used during hyperparameter tuning with GridSearchCV. The models were trained on an 80–20 train-test split. Performance was evaluated using R<sup>2</sup>, Mean Squared Error (MSE), and Mean Absolute Error (MAE). Among them, CatBoost consistently achieved the highest or joint-highest R<sup>2</sup> scores across all targets, while maintaining low prediction errors (Table 3). The best-performing configurations were as follows: for the PH model, learning rate = 0.05, max depth = 5, and n\_estimators = 300; for the MH model, learning rate = 0.05, max depth = 5, and n\_estimators = 300; and for the SH model, learning rate = 0.05, max depth = 5, and n\_estimators = 100.

Based on its superior performance and strong generalization capacity, CatBoost was selected as the primary modeling method. Further, to validate the robustness of CatBoost, a sensitivity analysis was conducted using three commonly applied random seeds (42, 77, and 2024). In each case, tenfold cross-validation and hyperparameter tuning were repeated. The R<sup>2</sup> scores remained stable across partitions, ranging from 0.635 to 0.755 (PH), 0.649 to 0.731 (MH), and 0.555 to 0.606 (SH), confirming the model's robustness. Detailed results are reported in Table 4.



### Data preprocessing

Data preprocessing involved encoding categorical control variables using target encoding, where categorical values were replaced by the mean dependent variable value for each category. Missing data were handled through imputation based on feature means. The dataset was standardized to ensure comparability across features.

Feature importance was initially assessed using CatBoost's built-in mechanism, which evaluates the reduction in prediction error attributed to individual features during tree splits. SHAP values were subsequently employed to validate and interpret feature importance rankings. This dual analysis enabled the identification of primary predictors for each health dimension.

In addition, descriptive statistics for all independent and dependent variables are presented in Table 5.

## Results

### Global analysis: feature contribution

SHAP analysis was used to evaluate the association of each feature on model predictions, providing a more detailed understanding of feature contributions compared to preliminary feature importance analysis. Two visualizations were used: the SHAP summary plot and the SHAP beeswarm plot. The summary plot ranks features by their mean absolute SHAP values, revealing the most influential features, while the beeswarm plot visualizes SHAP value distributions across samples, showing both feature importance and contribution direction. Points are color-coded (red for higher feature values, blue for lower values), with the x-axis showing the magnitude and variability of each feature's contribution [44]. The top ten most important features are presented in descending order of SHAP values.

### Physical health (PH)

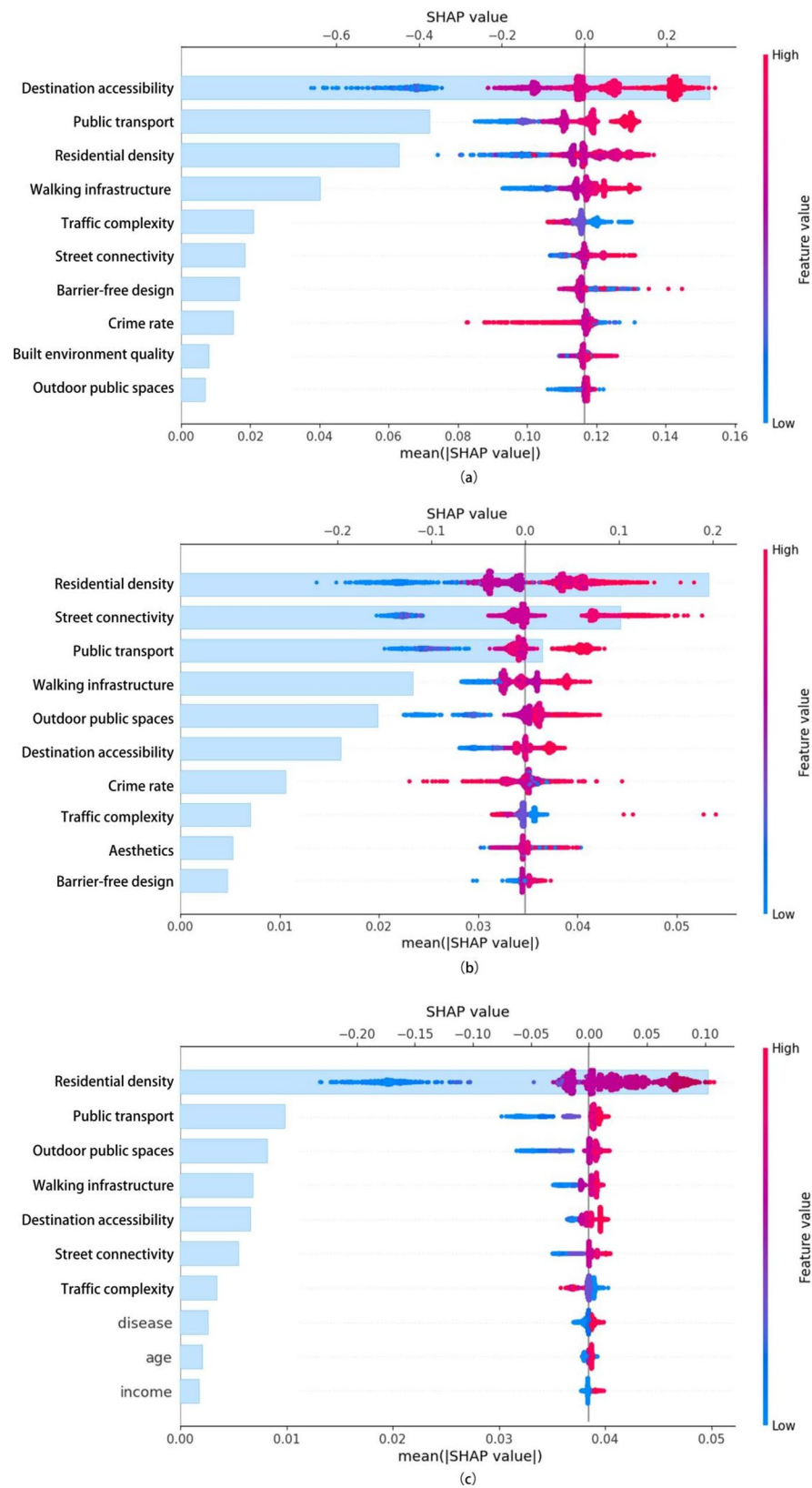
Figure 1(a) shows the SHAP summary and beeswarm plots for the PH model. The results indicate that destination accessibility, public transportation, residential density and walking infrastructure are the primary features positively associated with physical health—higher values of these variables correspond to higher predicted levels of physical health. In contrast, traffic complexity and crime rate exhibit strong negative associations, with higher variable values linked to lower SHAP values, suggesting potential inhibitory contributions on physical health. The associations of street connectivity, barrier-free design, outdoor public spaces, and built environment quality appear neutral, likely due to their relatively balanced distribution across the sample or limited variation in related contextual factors.

Destination accessibility stands out in the summary plot with a much higher importance compared to other features, and its positive association is also evident in the beeswarm plot. High destination accessibility allows residents to reach various essential services and public facilities more efficiently, encouraging daily physical activities, reducing travel time, and alleviating fatigue, which significantly benefits physical health. This aligns with existing research showing that residents in areas with high accessibility are more likely to engage in walking and cycling, activities with substantial health benefits [45]. Moreover, better accessibility reduces reliance on motor vehicles, thereby lowering health risks associated with sedentary behavior [46].

Public transportation and residential density also demonstrate substantial positive associations. Convenient public transit systems help relieve commuting stress and increase access to health services and recreational venues. Meanwhile, appropriately high-density residential

**Table 5** Descriptive statistics for independent and dependent variables

| Variable                  | Mean     | Std      | Min      | Max      | Range    |
|---------------------------|----------|----------|----------|----------|----------|
| PH                        | 2.323505 | 0.490007 | 0.65     | 3.05     | 2.4      |
| MH                        | 1.627783 | 0.26704  | 0.738047 | 2.592753 | 1.854707 |
| SH                        | 0.63741  | 0.163902 | 0        | 1        | 1        |
| Residential density       | 6.649725 | 1.385184 | 1.86     | 8.65     | 6.79     |
| Land-use diversity        | 2.485271 | 0.617235 | 1.588235 | 4.294118 | 2.705882 |
| Destination accessibility | 3.327884 | 0.770654 | 1        | 4        | 3        |
| Street connectivity       | 2.980861 | 0.665577 | 1        | 4        | 3        |
| Walking infrastructure    | 3.182915 | 0.696493 | 1        | 4        | 3        |
| Aesthetics                | 2.97807  | 0.632059 | 1        | 4        | 3        |
| Traffic complexity        | 2.283493 | 0.261671 | 1.666667 | 3.333333 | 1.666667 |
| Crime rate                | 2.985912 | 0.618723 | 1        | 4        | 3        |
| Public transport          | 3.263357 | 0.799083 | 1        | 4        | 3        |
| Outdoor public spaces     | 3.096624 | 0.676115 | 1        | 4        | 3        |
| Barrier-free design       | 3.024987 | 0.634789 | 1        | 4        | 3        |
| Green space               | 2.985885 | 0.612963 | 1        | 4        | 3        |
| Built environment quality | 3.072468 | 0.675263 | 1        | 4        | 3        |



**Fig. 1** SHAP plots of the PH/MH/SH models

areas tend to foster walkable environments, encouraging residents to engage in more frequent physical activity.

In contrast, traffic complexity and crime rate are strongly negatively associated with physical health. In particular, crime rate shows a concentration of red dots (indicating high crime rates) in the negative SHAP value range, suggesting that high perceived crime significantly undermines residents' health. This observation aligns with previous research showing that perceived safety is closely linked to the frequency of outdoor physical activity.

Overall, the functional (e.g., accessibility), supportive (e.g., walking infrastructure), and safety-related (e.g., low crime rate) attributes of the built environment collectively shape physical health outcomes. Promoting healthy aging thus requires a multi-dimensional and coordinated enhancement of these environmental features.

### **Mental health (MH)**

Figure 1(b) presents the SHAP summary plot for the MH model, illustrating the individual contributions of environmental features to predicted mental health outcomes. Overall, residential density, street connectivity, public transportation, walking infrastructure, outdoor public space, and destination accessibility exhibit positive associations with mental health. This is particularly evident when feature values are high (red points), as their SHAP values tend to cluster on the right-hand side of the plot, indicating stronger positive contributions. In contrast, traffic complexity and crime rate demonstrate negative associations, meaning that higher values correspond to more detrimental associations on mental health. Aesthetic quality and barrier-free design show relatively neutral association, with SHAP values centered near zero, suggesting weaker or less consistent contributions across the sample.

The positive association of residential density on older adults' mental health may be attributed to greater social interaction and community support systems. High residential density often coincides with more public facilities and services, which help older adults build stronger social connections, thereby improving mental health. Existing research has shown that moderate residential density can increase the frequency of community interactions, enhance social support, and improve psychological well-being [47]. In particular, for older adults, a robust social support network helps reduce loneliness, alleviate anxiety and depression, and enhance quality of life [48].

High street connectivity generally implies greater walkability and more accessible transportation networks, which encourage outdoor activity and spontaneous social encounters. Empirical research has linked street connectivity to improved mental health, particularly in urban environments, where better connectivity expands

residents' activity ranges, strengthens social participation, and fosters community cohesion [49]. In the current analysis, both street connectivity and public transportation demonstrate strong positive contributions, suggesting that well-connected transportation infrastructure may alleviate travel-related stress and inconvenience, thereby improving perceived psychological comfort. This trend is supported by the SHAP plot, where high feature values cluster predominantly on the right-hand side.

Of particular note is the role of crime rate, which emerges as a significant negative predictor. High crime rates correspond to substantially lower SHAP values, indicating persistent psychological stress and a sense of insecurity. Similarly, traffic complexity follows a comparable trend: low values (blue points) are concentrated on the right-hand side of the SHAP axis, whereas high values (red points) shift leftward, reinforcing the negative contribution of complex traffic conditions on mental well-being.

In summary, the MH model identifies key contributors to mental health that center around three primary dimensions: residential structure, transportation systems, and neighborhood safety. The findings highlight the importance of high-density, accessible, and safe urban environments in fostering resilient mental health support systems for older adults.

### **Social health (SH)**

Figure 1(c) shows the SHAP summary and beeswarm plots for the SH model. Overall, most variables have SHAP values concentrated around zero, indicating relatively weak contributions to the model output. Only a few variables exhibit moderate explanatory power.

Among the positively associated features, residential density demonstrates the strongest contribution. Its SHAP values range from negative to positive, indicating a phased pattern of association with social health: lower residential density (blue dots) is primarily associated with negative SHAP values, while higher residential density (red dots) corresponds to positive values. This suggests that high-density community environments may enhance neighborhood interactions and increase social engagement, thereby promoting social health.

Public transportation and outdoor public space also emerge as moderately influential positive factors. Higher levels of public transport convenience correspond to positive SHAP values, indicating that improved transportation access may facilitate social connectivity and participation, benefiting social health. Similarly, high-quality outdoor spaces can play a vital role in improving opportunities and quality of social interactions.

By contrast, negative factors are relatively weaker. Traffic complexity shows slight positive contributions at lower values, but its SHAP values decline at higher levels



(red dots), implying that increased traffic complexity may negatively affect social health under certain conditions, potentially due to perceived inconvenience or safety concerns.

Other variables such as walking infrastructure, destination accessibility, and street connectivity have SHAP values highly concentrated around zero, suggesting limited predictive power for social health after accounting for other factors.

In addition, although control variables (chronic disease status, age, and income) exhibit relatively weak SHAP values compared to environmental variables, they still demonstrate notable distributional patterns in the SH model. Specifically, income shows clear positive SHAP values at higher levels (red dots), suggesting that older adults with greater financial resources have more opportunities for social participation and thus better social health outcomes. Chronic disease status also shows a mild positive trend, with higher disease values (red dots) more frequently associated with positive SHAP contributions, indicating that some older adults with chronic conditions may engage in proactive social coping and self-regulation, thereby achieving improved social experiences. Age presents a more complex pattern: although the overall trend is weakly positive, some younger-old individuals (blue dots) also exhibit higher SHAP values, suggesting that the association between age and social health is not strictly linear and may be shaped by multiple interacting factors such as retirement status, physical capacity, and psychological adaptability.

In summary, the main positive contributors in the SH model are residential density, public transportation, and outdoor space, while traffic complexity emerges as a potential negative factor. In addition, the model reveals patterns associated with socioeconomic disparities, highlighting the multidimensional and intersecting nature of social well-being in aging populations.

#### Local analysis: feature interactions

While the global SHAP analysis identifies the overall importance and directionality of individual features across the dataset, local analysis focuses on how specific features contribute to predictions through their interactions with others at the individual sample level. This is achieved through SHAP dependence plots, which display how the SHAP value of one feature (x-axis) varies across its range, while the color gradient reflects the value of an interacting feature. This visualization reveals both the nonlinear relationship of a single variable and potential joint interaction with another, thereby uncovering subtle interaction patterns that global summaries may overlook [44].

This study focuses only on high-quality, interpretable interaction plots, rather than displaying all feature pairs.

Specifically, we retained only the SHAP dependence plots that met the following criteria: [1] a clear and nonlinear association pattern for the primary variable; [2] a consistent color gradient suggesting a potential interaction effect; [3] minimal noise or distortion from outliers.

#### Physical health (PH)

Figure 2 displays the interaction plots for the Physical Health (PH) model.

##### (1) Residential Density × Street Connectivity

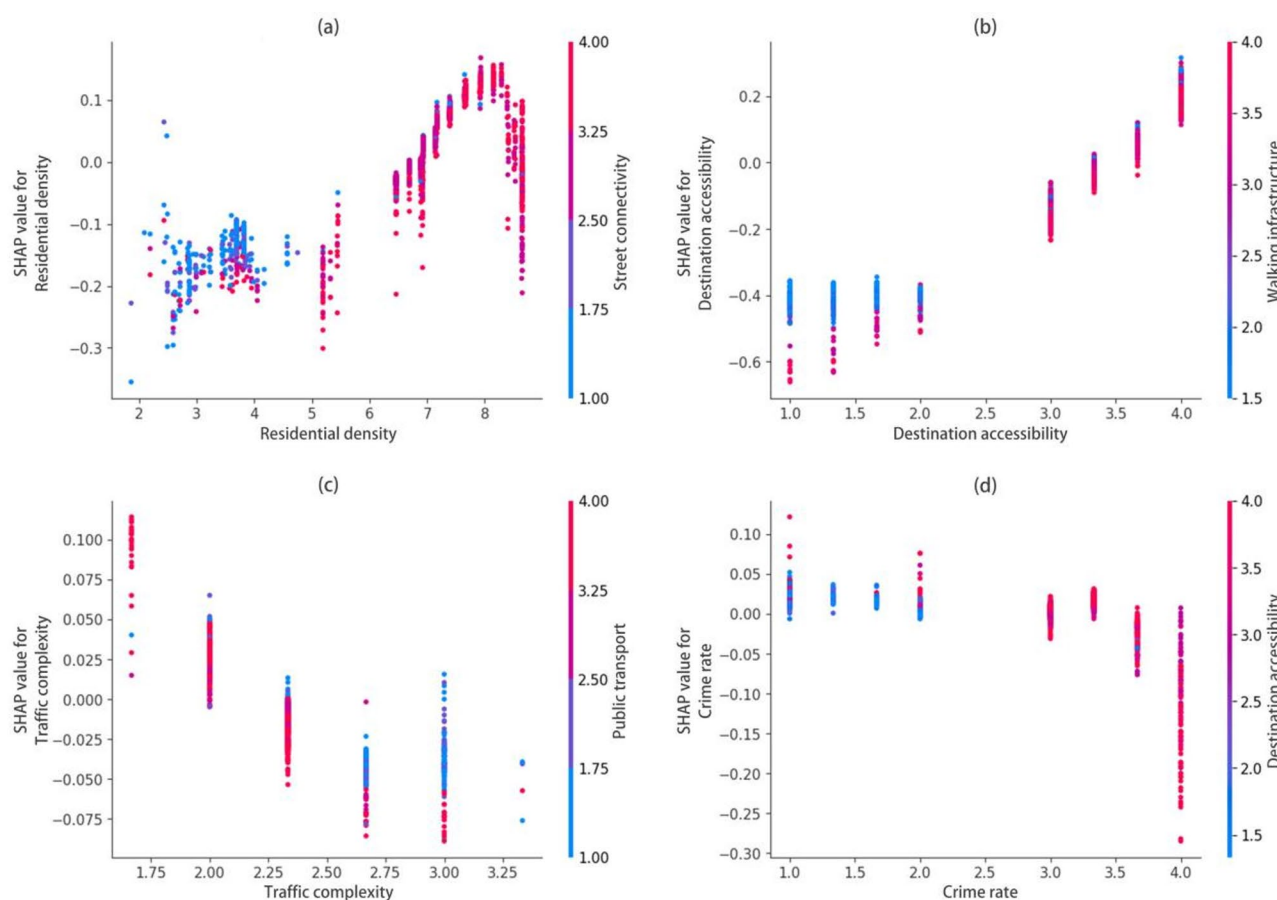
As shown in Fig. 2(a), residential density demonstrates a clear nonlinear association with physical health. In the low-density range (<5), SHAP values are mostly negative, indicating a weaker or even slightly unfavorable relationship with health outcomes. However, as density increases to a moderately high level ( $\approx 6$ –8), SHAP values rise noticeably, suggesting more favorable health-related associations—particularly in areas with high street connectivity (red dots). In contrast, in areas with low connectivity (blue dots), the positive pattern associated with increased density appears to be substantially reduced. These results imply that high-density development may be more supportive of health when accompanied by a well-connected street network.

##### (2) Destination Accessibility × Walking Infrastructure

Figure 2(b) presents the main contribution of destination accessibility and its interaction with walking infrastructure. Overall, destination accessibility is positively associated with physical health, with SHAP values increasing as destination accessibility rises. A closer look shows that the positive association between destination accessibility and physical health is more pronounced when walking infrastructure is well-developed (red dots), whereas this association appears weaker in areas with insufficient walking facilities (blue dots). This indicates that accessibility alone is insufficient; it must be accompanied by supportive walking infrastructure to promote physical health.

##### (3) Traffic Complexity × Public Transportation

The negative relationship between traffic complexity and physical health shows a nuanced pattern under different conditions (Fig. 2c). When traffic complexity is low (<2.5), the associated adverse patterns appear partially attenuated in areas with good public transportation (red dots), suggesting that efficient transit systems may be linked to reduced health risks in traffic-heavy environments. However, in high traffic complexity zones (>2.5), even a high level of public transit availability fails to alleviate—and may even exacerbate—the negative association. This “reverse moderation” relationship may stem from the excessive burden on the overall transport system, indicating that simply increasing public transit resources may be insufficient in high-pressure areas.



**Fig. 2** Significant Dependence Plots for PH

Instead, a systemic optimization of travel patterns and urban experience is needed.

#### (4) Crime Rate × Destination Accessibility

Figure 2(d) shows a clear negative relationship between crime rate and physical health. In low-crime areas, the association is relatively weak or even slightly positive; however, in high-crime environments ( $\geq 3$ ), SHAP values drop sharply, indicating a significantly intensified negative association. Further analysis reveals that even under conditions of high accessibility (red dots), high crime rates can severely suppress the otherwise positive contribution. This interaction highlights that public safety is a foundational prerequisite for health-oriented urban design.

#### Mental health (MH)

Figure 3 displays the interaction plots for the Mental Health (MH) model.

##### (1) Residential Density × Crime Rate

As shown in Fig. 3(a), the SHAP values of residential density exhibit a clear upward trend with increasing values, indicating that moderately high-density communities ( $> 5$ ) are generally beneficial for mental health.

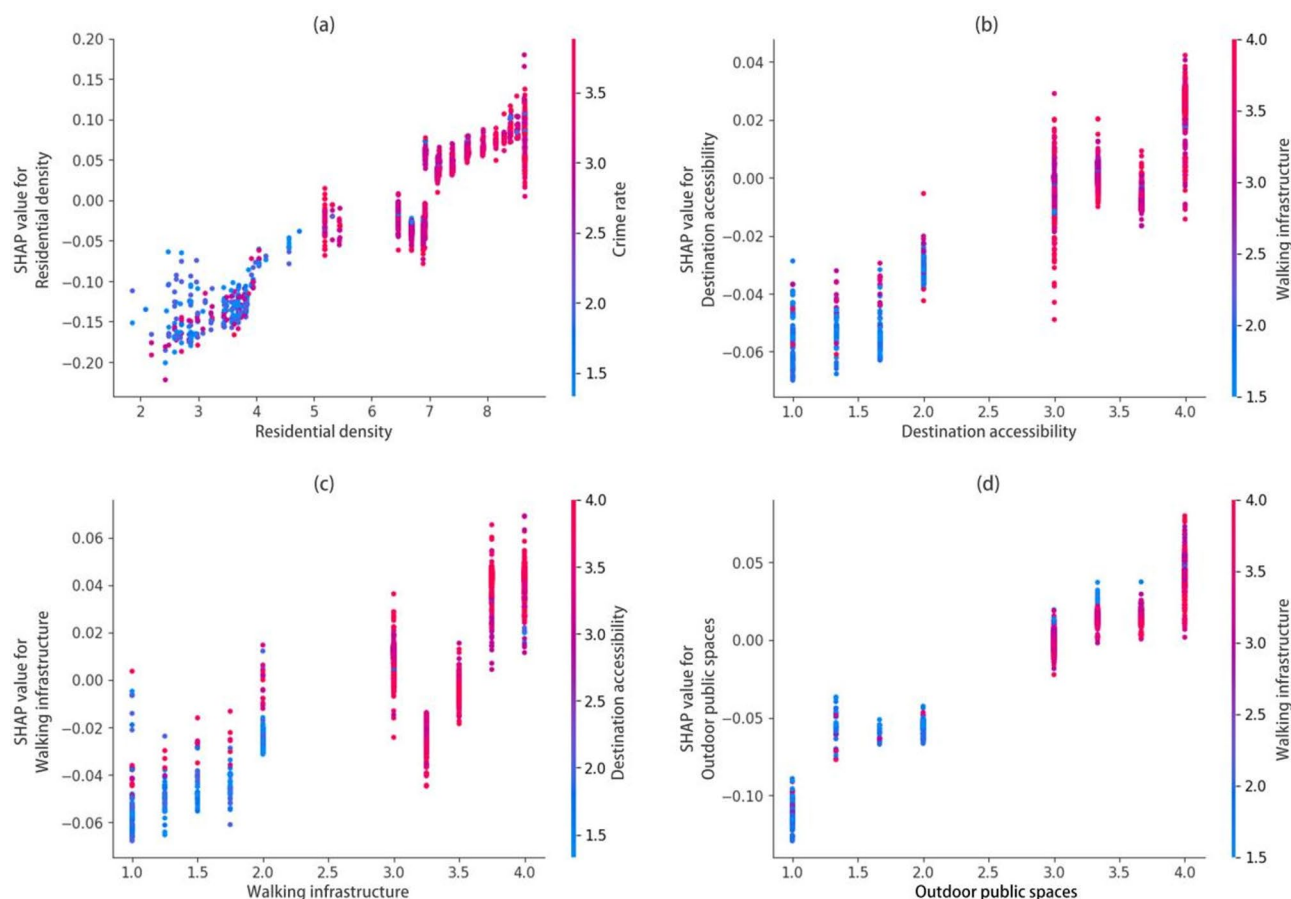
However, this positive association is notably weakened in areas with high crime rates. In the plot, red points (indicating high crime rate) in high-density areas correspond to lower SHAP values, while blue points (low crime) align with higher SHAP contributions. This suggests that residential density alone does not guarantee improved mental health—its benefits are contingent upon a safe neighborhood environment.

##### (2) Destination Accessibility × Walking Infrastructure

Figure 3(b) shows that destination accessibility has a consistently positive association on mental health. It is further amplified when walking infrastructure is well-developed, as indicated by the red dots clustering on the right side of the SHAP axis. This synergy implies that neighborhoods with both high accessibility and supportive walking conditions are better equipped to encourage daily activities and social participation among older adults, thus improving mental well-being.

##### (3) Walking Infrastructure × Destination Accessibility

Figure 3(c) further supports the above findings through the interaction between walking infrastructure and destination accessibility. When walking infrastructure is very poor ( $< 2$ ), its contribution to mental health is almost



**Fig. 3** Significant dependence plots for MH

entirely negative. However, as walking infrastructure improves, SHAP values rise significantly, particularly in areas with high destination accessibility. In other words, high-quality walking environments only reach their full potential when paired with accessible destinations, and the two dimensions must be optimized together to achieve the best outcomes.

#### (4) Outdoor Public Space × Walking Infrastructure

Figure 3(d) shows that the association of on mental health follows a clear stage-like pattern. When outdoor public space is low (<2.5), SHAP values are mostly negative, suggesting that poor or missing outdoor spaces may contribute to psychological distress. However, once outdoor public space exceeds 3.0, its positive contribution becomes more apparent, especially in areas with well-developed walking infrastructure. This finding underscores the importance of combining “usability” (space quality) with “accessibility” (infrastructure): improving one dimension alone is insufficient, and a systemic co-optimization is required.

#### Social health (SH)

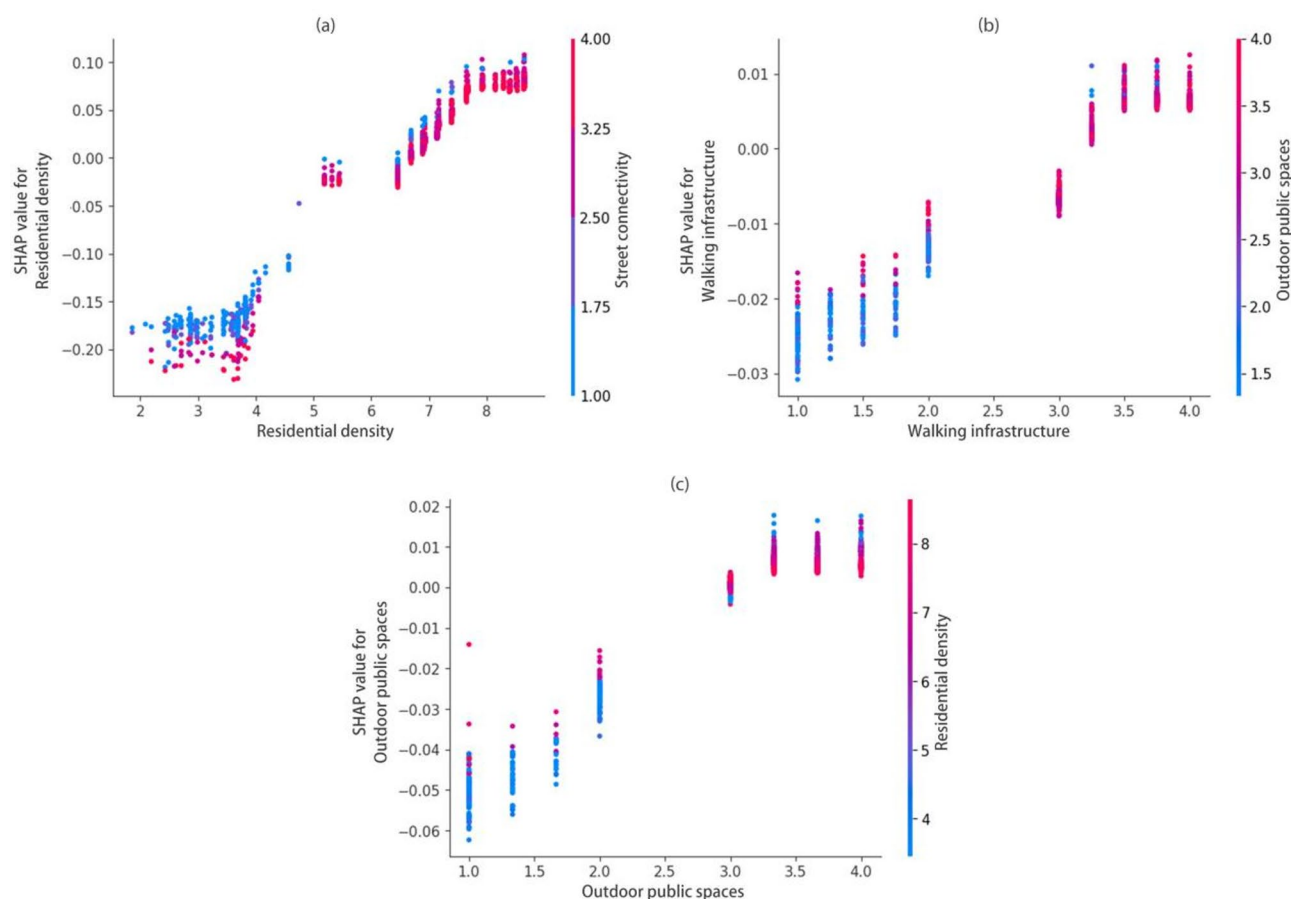
Figure 4 displays the interaction plots for the Social Health (SH) model.

##### (1) Residential Density × Street Connectivity

Figure 4(a) shows that the association of residential density on social health is markedly nonlinear. In low-density areas (<5), SHAP values are mostly negative, indicating that sparse populations may reduce the frequency of community interaction and utilization of public services, thereby weakening social health. As residential density increases to a moderate-to-high level (>6), SHAP values rise significantly, suggesting a positive association—especially when street connectivity is high. This highlights that high-density residential environments must be accompanied by well-connected street networks to foster social engagement and cohesion, as the two work synergistically to enhance social health.

##### (2) Walking Infrastructure × Outdoor Public Space

Figure 4(b) illustrates a nonlinear pattern in the relationship between walking infrastructure and social health. When walking infrastructure is rated low (<3), SHAP values are generally negative, indicating that limited walkability tends to be associated with reduced



**Fig. 4** Significant dependence plots for SH

mobility and lower levels of social participation. However, when walking infrastructure improves ( $\geq 3$ ), a positive association emerges—particularly in areas where outdoor public space quality is also high. This pattern suggests a potential synergistic relationship between walkability and the availability of public gathering spaces: street improvements alone may not correspond with higher social health scores unless accompanied by accessible and inviting venues for social engagement.

### (3) Outdoor Public Space $\times$ Residential Density

In Fig. 4(c), outdoor public space displays a staged interaction pattern with residential density. When outdoor public space is low ( $< 2.5$ ), SHAP values are clearly negative, indicating that a lack of public space can erode community cohesion. As outdoor public space improves to a moderate or high level ( $\geq 3$ ), SHAP values gradually shift into the positive range, reflecting the beneficial role of high-quality outdoor environments in promoting social health. This positive relationship is especially pronounced in high-density neighborhoods. The result underscores that in densely populated cities, public spaces must not only be present but also of high quality to fully realize their social function.

## Discussion

This study leveraged CatBoost models and SHAP interpretability to examine how built community environments relate to the physical, mental, and social health of older adults through complex feature contributions and interactions. Compared with traditional linear models, SHAP offers a more refined understanding of multi-dimensional health patterns, particularly by capturing nonlinear, context-dependent, and synergistic associations—thereby providing insights with greater interpretive depth and practical relevance for healthy urban development.

### General findings

#### *Physical health: dual support from functional provision and risk barriers*

The results show that destination accessibility, public transportation, residential density, and walking infrastructure are key facilitators of physical health. These features together form a “high-accessibility, high-mobility” support system that lowers travel thresholds, encourages physical activity, and enhances the functional efficiency of daily life for older adults. This aligns with previous



studies suggesting that greater accessibility reduces travel costs and promotes physical and flexible daily routines in older adults [28]; public transit fosters “active commuting” through incidental walking to transit stops [7]; and moderate residential density supports walkable neighborhoods and familiar social networks [25], leading to higher levels of physical engagement.

However, crime rate and traffic complexity act as major deterrents. Their adverse associations extend beyond direct links to stress and mobility—they also appear to attenuate the contributions of features that are generally considered beneficial. Studies indicate that high traffic burden and perceived insecurity reduce the frequency of public space use [31], while also increasing risks of accidents, pollution exposure, and injuries. Thus, it is not enough to improve infrastructure supply alone; efficient health promotion must also build residents’ “confidence to use” environments through risk mitigation strategies such as traffic regulation and public safety governance.

Particularly noteworthy are the interdependencies among variables. The observed interaction between destination accessibility and walking infrastructure indicates that accessibility, in the absence of adequate pedestrian support—what might be termed “floating accessibility”—may not translate into meaningful health advantages. Similarly, the combined presence of traffic complexity and crime appears to offset the potential benefits of otherwise favorable features. These patterns suggest that functional infrastructure, if not accompanied by a safe surrounding environment, may contribute to feelings of anxiety, avoidance, or withdrawal, potentially limiting its effectiveness. This highlights the importance of environmental safety as a foundational condition for promoting health-supportive settings.

#### ***Mental health: synergy between social potential and restorative environments***

The mechanisms underlying mental health improvements depend more heavily on social interaction opportunities and emotional support structures. Residential density and street connectivity provide more than physical access—they create “structures of social opportunity”: high population concentration, frequent movement, and vibrant neighborhood contact collectively enhance emotional support, belonging, and identity among older adults [47, 48]. This integration of social capital and spatial design helps explain why moderately dense, highly connected environments are often associated with lower levels of loneliness and anxiety [49].

Meanwhile, outdoor public spaces not only serve as social platforms but also function as restorative environments. Interaction plots show that high-quality space alone is insufficient; it must be coupled with a supportive walking network to fully activate its psychological

benefits. This aligns with the “restorative environment theory,” which emphasizes the joint importance of usability and accessibility.

On the negative side, traffic complexity and crime remain key psychological stressors. Prior studies note that complex traffic conditions and safety concerns not only cause acute anxiety but also accumulate into chronic stress over time [36]. Therefore, improving mental health requires both “enabling participation” and “removing the fear of mobility.”

#### ***Social health: Spatial structuring and the activation of interaction potential***

Compared with physical and mental health, the pathway to social health is more “dependence-driven” and requires a triple match of “people–space–interaction.” Neither accessibility nor density alone guarantees robust social connection; spatial layouts must also offer explicit “spaces for socialization.” Residential density and public transportation provide a baseline frequency of potential encounters, while high-quality outdoor spaces offer the actual venues for interaction. These findings are consistent with research showing that higher density increases the probability of contact, and public spaces enable community interaction, thereby fostering cohesion and belonging [24, 25].

It is notable that most accessibility-related variables show relatively weak associations in the SH model, suggesting that “being able to reach” does not necessarily equate to “wanting to stay.” Interaction plots further reveal that combinations of outdoor spaces with residential density and walking infrastructure are associated with stronger effects: high-density areas lacking social spaces and mobility support may correspond to “group isolation,” while low-density neighborhoods—even with ample public spaces—may still not foster meaningful interaction. This pattern aligns with the “optimal density hypothesis” and supports the notion of an inverted U-shaped relationship [24–26]: too low a density may restrict opportunities for social contact, whereas excessively high density may correspond to crowding, conflict, and social withdrawal—potentially undermining social well-being.

#### ***The nonlinear and multidimensional role of residential density***

Residential density exhibits a particularly complex and prominent pattern across all three dimensions of health. SHAP dependence plots indicate that when residential density is low ( $<5$ ), its associations with physical, mental, and social health tend to be weak or even negative, suggesting limited or potentially unfavorable conditions for overall well-being. In contrast, when density increases to a moderately high range ( $\approx 6$ – $8$ ), SHAP values rise



substantially, implying that such environments may better support social interaction, service accessibility, and walkability—factors commonly linked with improved multidimensional health outcomes. Notably, when density exceeds approximately 8.5, SHAP values for physical health show a slight decline, potentially reflecting challenges associated with overconcentration, such as spatial congestion, noise, and perceived environmental stress.

This finding strongly supports the theoretical hypothesis of an “inverted U-shaped relationship” between residential density and health. That is, the benefits of density are not linearly incremental but peak within an optimal range. This conclusion aligns well with theories in environmental psychology and social capital: moderate density can enhance neighborhood interaction and strengthen social support networks, thereby reducing loneliness and the risk of depression [25]; conversely, excessively high density may lead to loss of privacy, increased exposure to air and noise pollution, and greater physical and emotional discomfort [26]. Previous studies have also noted that while a certain level of concentration improves service efficiency and urban vitality, over-density may reduce overall quality of life [23].

More importantly, interaction analyses show that the health associations of residential density are not absolute but contingent on supportive environmental conditions. The positive contribution of moderate density are most pronounced in settings with high levels of destination accessibility, public transportation availability, pedestrian infrastructure, and outdoor public space. This highlights the ecological nature of health determinants—no single factor (such as density) can independently drive health outcomes without coordination with other environmental components. In areas lacking facilities or with poor transportation access, even moderate density may fail to yield health benefits and may, in fact, exacerbate negative exposures.

### **The role of socioeconomic and demographic characteristics**

While environmental features play a central role in shaping older adults’ social health, the SHAP analysis also highlights the non-negligible contributions of socioeconomic and demographic control variables—particularly income, age, and chronic disease status. Although these variables exhibit relatively modest SHAP values compared to key environmental predictors, their inclusion as covariates reveals underlying inequalities in how social health is distributed across population groups.

Among them, income shows a consistently positive association with social health, indicating that individuals with higher socioeconomic resources may have better access to social participation opportunities, greater mobility, and stronger support networks. This aligns

with evidence suggesting that higher income enhances social integration by enabling older adults to afford participation in community events, maintain broader social networks, and access diverse social venues [50, 51]. Conversely, chronic disease status, though typically associated with reduced physical functioning, may not necessarily hinder social well-being. Existing research indicates that individuals with chronic illnesses often engage in compensatory behaviors, such as community involvement and peer support, to manage their health and maintain social connectedness—an adaptive strategy commonly observed in self-management practices [52, 53].

Age, meanwhile, presents a more nuanced picture. Although aging is often accompanied by retirement and increased discretionary time that may be conducive to social participation [50], it can also bring about social losses, such as bereavement, caregiving burdens, or physical decline, which may diminish social engagement. The SHAP results reflect this duality, with both positive and negative contributions depending on the individual’s life context. This highlights the importance of age-sensitive interventions that account for heterogeneity in later life experiences [54, 55].

Collectively, these findings support a broader understanding of social health as an outcome shaped by both environmental and structural factors. This aligns with the theory of environmental deprivation, which posits that disadvantaged populations—such as those with lower income or poor health—may experience reduced access to social opportunities due to spatial, physical, or systemic barriers [56, 57]. Merely improving the physical environment may not be sufficient if vulnerable groups are structurally excluded from utilizing such improvements.

Therefore, policy interventions should not only enhance the built environment but also prioritize inclusive planning strategies that target at-risk populations. These may include subsidized access to community facilities, chronic disease-friendly social programs, and tailored outreach for older adults with mobility or resource constraints. Additionally, redistributive policies that ensure equitable provision of walkable, service-rich neighborhoods can help narrow social health disparities.

In sum, the interplay between individual resources and community structures must be addressed in tandem. A truly inclusive, health-supportive community environment requires integrated efforts that target both material inequalities and social vulnerability, ensuring that the benefits of community design are equitably shared across all segments of the older adult population.

### Interactions and synergistic mechanisms of environmental factors

The SHAP interaction analysis identified multiple key pairs of environmental features that were jointly associated with older adults' health outcomes. However, these interaction patterns were not consistently observed across the three health dimensions—physical, mental, and social—but instead varied in form and prominence, highlighting the complexity and domain-specific nature of how environmental characteristics relate to health [12].

Residential density emerged as the most recurrent and prominent interaction partner across all three models, serving as a structural baseline whose associations with health outcomes varied depending on contextual factors. In the case of physical health, stronger associations were observed in areas with high street connectivity; for social health, residential density was more strongly related to outcomes when ample outdoor public spaces were present; and for mental health, the relationship appeared to be notably shaped by local crime rates [58]. This suggests that residential density functions as an “amplifier” rather than a standalone determinant—its potential can only be realized through synergy with other urban features [59].

The combination of destination accessibility and walking infrastructure demonstrated consistent positive interactions in both the physical and mental health models. In both cases, high accessibility alone was insufficient—only when supported by well-developed walking environments did its full health-promoting potential emerge [60]. This finding supports integrated urban design principles that emphasize the joint importance of accessibility and usability, particularly for older adults with mobility constraints.

In contrast, interaction mechanisms in the social health model leaned more heavily toward spatial quality and spatial organization. Interactions such as those between outdoor public spaces and walking infrastructure, or between outdoor spaces and residential density, emphasized that social engagement depends not just on the provision of public space, but on its accessibility, functionality, and experiential quality [61]. These findings highlight a domain-specific logic for social health, which is distinct from the functional emphasis in physical health and the perceptual/safety-driven nature of mental health.

It is also noteworthy that some interaction patterns appeared prominently only within specific health domains. For instance, the negative interaction between crime rate and destination accessibility was prominent in the physical health model, likely due to the higher exposure involved in outdoor activities. However, this pattern was not observed in the social health model, reflecting varying degrees of vulnerability and behavioral response

across health dimensions. Such discrepancies support ecological health frameworks that emphasize person–environment–task fit as a dynamic and context-dependent determinant of health.

Taken together, these findings underscore the presence of key interaction patterns and the heterogeneous ways in which environmental features relate to distinct health domains. For policy and planning, this implies that built environment interventions should be tailored to specific health goals rather than adopting a one-size-fits-all approach. To promote physical health, priority should be given to enhancing access and traffic safety; for mental health, the focus should be on reducing crime and improving the usability of green spaces; and for social health, investments should be made in creating high-quality public spaces in medium- to high-density neighborhoods. A systemic, interaction-aware perspective on urban design is essential for achieving synergistic improvements in the multidimensional health of aging populations.

### Limitations and future research

This study leverages CatBoost and SHAP to advance the analytical framework for multidimensional health prediction. The interpretability of SHAP improves the transparency of machine learning models, enhancing their utility in public health research. However, several limitations merit attention.

First, this study employs a cross-sectional survey design, which inherently limits the ability to draw causal conclusions. Although the CatBoost model, paired with SHAP, allows us to uncover complex, nonlinear associations and interaction effects between environmental features and health outcomes, the observed relationships should be interpreted as correlational rather than causal. Temporal precedence—the essential foundation for causal inference—cannot be established in the current design, as both independent and dependent variables were measured simultaneously. Additionally, the possibility of reverse causality (e.g., healthier individuals being more likely to choose better environments) cannot be ruled out. To strengthen causal claims, future studies should consider longitudinal or panel data that track changes in health status and environmental exposure over time. This would help disentangle temporal dynamics and better isolate the directional pathways through which built environments shape multidimensional health outcomes in aging populations.

Second, while SHAP enhances model interpretability, it remains a predictive tool and does not account for unobserved confounders. To address this limitation, future studies could incorporate causal inference approaches—such as instrumental variable analysis, propensity score methods, or structural equation modeling—to better

isolate the causal association of environmental features on health outcomes. Combining interpretable machine learning with causal techniques may offer a more rigorous framework for informing public health interventions.

Third, although the current analysis was conducted at the city-wide level, the sample covers seven administrative districts across Shanghai's inner, middle, and outer rings, encompassing diverse urban forms and demographic structures. While we did not conduct separate models for each district, observational evidence suggests that the health implications of specific environmental features may differ across spatial contexts—for example, between older, high-density neighborhoods such as Hongkou and Yangpu, and newer urbanizing districts such as Jiading and Songjiang. Future studies could investigate such regional heterogeneity through district-level stratified analysis or spatially explicit modeling, offering more tailored guidance for place-sensitive planning and interventions.

## Conclusion

This study employed CatBoost and SHAP to examine how community public space features are associated with older adults' physical, mental, and social health. Key positive contributors included residential density, destination accessibility, and public transportation, while crime rate and traffic complexity were consistently negative. The results underscore the importance of interaction effects—environmental features must work in synergy to support healthy aging, as no single element (e.g., density or access) suffices in isolation. These insights inform urban planning by highlighting the value of medium-density neighborhoods (density index  $\approx 6$ –8) equipped with walkable infrastructure, accessible public transport, and high-quality public spaces. In parallel, reducing environmental stressors such as crime and traffic complexity is vital. Lastly, inclusive design and targeted support for vulnerable groups can ensure that built environment improvements translate into equitable health gains across the aging population.

## Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12889-025-23402-y>.

Supplementary Material 1

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## Author contributions

MZ and TS contributed to the conceptualization, methodology, investigation, and data curation of the study. YL conducted the literature review and questionnaire collection, while QQL was responsible for visualization.

YQL contributed to the conceptualization, methodology, and supervision of the study. All authors read and approved the final manuscript.

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## Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

## Declarations

### Ethics approval and consent to participate

This study was approved by the Science and Technology Ethics Committee of Tongji University, Shanghai, China (approval no. Tjdxsr2024039). All procedures were performed in compliance with relevant laws, institutional guidelines, and the ethical standards of the Declaration of Helsinki. No personal identifying information was collected, and informed consent was obtained from all participants involved in the study.

### Consent for publication

Not applicable.

### Competing interests

The authors declare no competing interests.

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